Varunkrishna K

ms23btech11033@iith.ac.in | k.varun0627@gmail.com | +91 7010887308 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.24
JAWAHAR HR SEC SCHOOL NEYVELI	Class XII (CBSE)	2023	83.00%
JAWAHAR HR SEC SCHOOL NEYVELI	Class X (CBSE)	2021	86.80%

SKILLS

Languages: C++, Java

Frameworks: Spring Boot, Hibernate/JPA

Databases: PostgreSQL, DynamoDB , Redis

Testing: JUnit, Mockito, Integration Testing

DevOps & Tools: Docker, Maven, Postman

EXPERIENCE

Kotak Mahindra Bank

Backend Engineering Intern, Cross Border Payment Hub Team

Tech Stack: Java, Spring Boot, DynamoDB, Resilience4j, Mockito, JUnit, Podman

- Designed and implemented an **Idempotent and Race-free** Payment Identity generation service , using a caller supplied **idempotent keys** and database level **atomic conditional writes** , ensuring consistent identity generation and preventing duplicate PaymentIDs during retries and concurrent requests.
- Utilized **SOLID principles , System Design Patterns** and used DynamoDB **single-table schema** to efficiently manage payment identities and idempotency records.
- Achieved **80%+ code coverage** across other production microservices, by writing **Unit and Integration Test** using **Mockito and JUnit** , validating payment workflows and service integrations and improving production readiness.

PROJECTS

Loan Management Microservices Platform | GitHub :MoneyLending, EurekaServer, kyc-service, notification-service

Tech Stack: Java, Spring Boot, Hibernate/JPA, PostgreSQL, Eureka, OpenFeign, Resilience4j, Caffeine

- Architected a **3-service microservices** loan platform (**Money Lending, KYC Verification, Notification**) with **Eureka**-based service discovery, **OpenFeign** for inter-service communication, and **Resilience4j** circuit breakers with fallback mechanisms for fault-tolerant availability.
- Ensured financial data consistency under concurrent access by combining **PostgreSQL optimistic locking** for high-throughput operations with **JPA pessimistic locking** for critical transactions.
- Preserved audit trail integrity independent of business transaction outcomes through **transaction propagation**, backed by centralized **Spring AOP aspects** for audit logging and performance monitoring.
- Applied **SOLID principles** with **System Design Patterns** for better scalability and testability; used **scheduled async thread pools** for overdue-loan processing with notification dispatch, and **Caffeine caching** to reduce database load on frequently accessed data.

Online Judge | GitHub : Online-Judge

Tech Stack : Java, Spring boot, Redis, PostgreSQL , Docker

- **Designed a Online Judge platform** with 2-service microservices (**Problem service and Code Execution Service**), communicating asynchronously rather than via tight coupling.
- **Applied SOLID principles and System Design Patterns** to support pluggable languages (C++/Java/Python) and

an extensible verdict generator for the submitted code to ensure scalable and maintainable code.
- **Built a crash-resilient async execution pipeline using Redis Streams with consumer groups**, implementing XCLAIM-based pending-message recovery so in-flight submissions survive a service restart without being lost.
- **Containerized the application with Docker and Docker Compose** for better security from malicious code submission and better isolation for other services.

Mini TCP HTTP Server & Todo Application | GitHub : Mini-Java-Server

Tech Stack : Java, JDBC, PostgreSQL, Jackson, Maven

- Built a **multithreaded HTTP server** using **Java ServerSocket** and thread pools, enabling concurrent request processing.
- Built a custom **JDBC connection pool** using BlockingQueue to reuse connections and bound concurrent database access.
- Built a **RESTful Todo API** with Controller-Service-Repository architecture and Jackson, supporting CRUD operations with PostgreSQL for persistence.
- Implemented **graceful shutdown** by closing the TCP server, terminating worker threads, and releasing pooled database connections.

SCHOLASTIC ACHIEVEMENTS

Achieved **Knight Rank** on LeetCode with a current rating of **1854** and a Peak Rating of **1900 (Top 5% globally)** and solved **1000+** problems across different topic with varying difficulties. LeetCode Profile

Secured **6 stars** in HackerRank problem solving . HackerRank Profile

RELEVANT COURSES

Data Structures and Algorithms, Database Management Systems-I, Introduction to C programming

EXTRACURRICULAR

- Regular gym going person with particular interest in powerlifting and strength trainings.
- Occasionally share technical learnings and insights on LinkedIn.